

Answer **all** the questions.

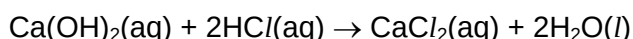
For
Examiner's
Use

1 Planning (P)

One method of determining the solubility product (K_{sp}) of a sparingly soluble salt is to dissolve it in water to form a saturated solution and titrating it against a known concentration of a suitable reagent at laboratory temperature.

A student was given a solution, **FA1**, that contains a saturated solution of calcium hydroxide, Ca(OH)_2 , a sparingly soluble salt at laboratory temperature. She was asked to determine the solubility product of calcium hydroxide by titrating **FA1** against a standard solution of hydrochloric acid.

Calcium hydroxide undergoes neutralisation reaction with hydrochloric acid as shown in the following equation.



In a preliminary test, she found that 25.0 cm^3 of **FA1** required around 4.5 cm^3 of $0.200 \text{ mol dm}^{-3}$ of a stock solution of hydrochloric acid for complete reaction. She decided that the concentration of this solution was too high and went on to prepare a diluted solution of hydrochloric acid, **FA2**, for titration against the **FA1** solution.

- (a) Write an expression for the solubility product of calcium hydroxide, stating its units.

[1]

- (b) (i) Given that the error of each measurement using the burette is $\pm 0.05 \text{ cm}^3$, calculate the percentage error of the titre volume obtained in the preliminary test and hence explain why the stock solution of hydrochloric acid was unsuitable for use in the student's titration.

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- (ii) Calculate an appropriate concentration of hydrochloric acid in the **FA2** solution to be used for the titration. Hence, calculate the volume of the stock solution required to prepare 250 cm^3 of this solution.

[3]

You may assume that you are provided with:

- The colours of the three indicators in acidic and alkaline solutions are given below:

Indicator	Colour	
	acidic solution	alkaline solution
methyl orange	red	yellow
brothymol blue	yellow	blue
cresol red	yellow	reddish purple

- details for the preparation of 250.0 cm³ of **FA2** solution from the stock solution of hydrochloric acid;
- details of the titration procedure;
- the choice of indicator, and the colour change at the end-point of the titration.

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[5]

- (d) In the student's titration, she found that 25.0 cm^3 of **FA1** reacted with exactly $y \text{ cm}^3$ of **FA2** of concentration $M \text{ mol dm}^{-3}$.

Outline how you would use her results to determine:

- (i) the concentration of Ca(OH)_2 in **FA1**;

- (ii) the solubility product of Ca(OH)_2 .

[2]

- (e) Identify **one** potential safety hazard in this experiment and state how you would minimise this risk

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[1]

[Total: 12 marks]

- 2 Water chemistry plays an important role in aquariums. For disinfection purposes, chlorine, Cl_2 , and chloroamine, NH_2Cl , are introduced into our tap water to destroy bacteria and viruses. However, these two chlorine-containing compounds are extremely harmful to fishes as they cause gill burns and blood poisoning.

Some water parameters found in our tap water are listed below.

chemical parameters	average reading / mg dm^{-3}	average reading / mol dm^{-3}
$\text{Cl}_2(\text{aq})$	2.3	
$\text{NH}_2\text{Cl}(\text{aq})$	2.0	
PO_4^{3-}	0.012	
Fe (as Fe^{2+} & Fe^{3+})	0.006	

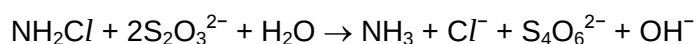
Table 1

Use of the information in Table 1 is relevant to questions (a) – (d).

- (a) Complete Table 1 by calculating the average reading of each parameter in mol dm^{-3} . [1]

- (b) During a water change, a water conditioner containing sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$, is added to the tank to remove both the aqueous chlorine and chloroamine present in tap water. A rectangular tank, with dimensions of 4 feet by 2 feet by 2 feet, typically can hold 360 dm^3 of water.

The balanced equation for the reaction that occurs between chloroamine and thiosulfate ion is given below:



- (i) Write an equation for the reaction that occurs between chlorine and the thiosulfate ion.

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- (ii) Calculate the volume of a $0.010 \text{ mol dm}^{-3}$ sodium thiosulfate solution that would be required for a 50% water change in the tank.

[3]

- (c) One of the commonly encountered problems in aquariums is algae bloom, which is caused by an excess of phosphates in the tank water. To reduce the phosphate concentration, it has been suggested that calcium supplement tablets could be added to precipitate out solid $\text{Ca}_3(\text{PO}_4)_2$.

Given that the numerical value of the solubility product, K_{sp} , of calcium phosphate is 2.07×10^{-33} and using the information given in **Table 1**, determine the minimum concentration of Ca^{2+} required for precipitation to occur in our tap water.

[2]

- (d) Our tap water contains trace amounts of iron, a typical transition element that forms iron(II) and iron(III) ions. These aqueous ions have colours in the green-yellow part of the visible light spectrum.

- (i) Complete the electronic configuration of iron atoms and of iron(II) ions.

Fe: $1s^2 2s^2 2p^6$

Fe^{2+} : $1s^2 2s^2 2p^6$

- (ii) Explain why both aqueous iron(II) and iron(III) ions are coloured.

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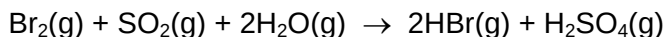
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[4]

- (e) Seawater contains a low concentration of bromide ions. Bromine is commercially extracted by the following series of steps.

I Chlorine is passed into acidified seawater and bromine is then removed in a stream of air. The concentration of bromine in the gas phase is too low at this stage for it to be condensed efficiently.

II Thereafter, the bromine produced is converted to hydrogen bromide by adding sulfur dioxide and a small excess of water vapour.



A moderately concentrated aqueous mixture of hydrobromic and sulfuric acids is obtained and the vapour is cooled and condensed.

III The concentrated mixture of acids is treated with chlorine and steam, producing a vapour from which liquid bromine is condensed on cooling. The sulfuric acid remains in the solution.

IV The crude bromine is purified by fractional distillation.

- (i) With reference to step **I** of the extraction, state and explain the relative oxidising power of the two halogens, chlorine and bromine.

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- (ii) Complete the table below, by giving the colours of chlorine and bromine when dissolved in water and in hexane.

solvents	chlorine, Cl_2	bromine, Br_2
water		
hexane		

- (iii) By considering the volatility of the two halogens, suggest the purpose of introducing steam into the mixture in step **III**.

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- (iv) Hence explain why, in step **III**, H_2SO_4 remains in the solution.

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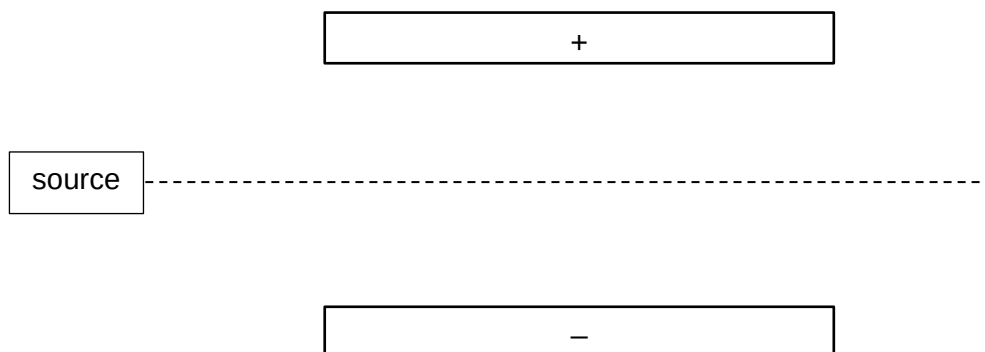
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[6]

[Total: 16 marks]

- 3 (a) Metals can be ionised by passing a large current through a metal filament.

In the diagram below, sketch and label the paths of Na^+ and Al^{3+} ions as they pass through an electric field.



[2]

- (b) Explain why the second ionisation energy of aluminium is:

- (i) higher than its first ionisation energy;

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- (ii) higher than the second ionisation energy of silicon.

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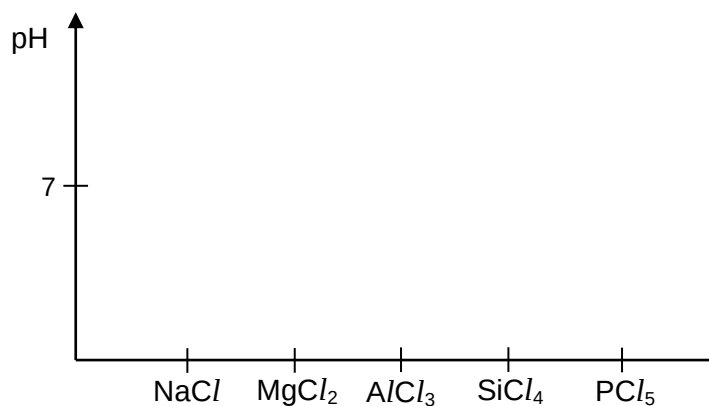
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[3]

- (c) The chlorides of Period 3 elements, sodium to phosphorus, dissolve in or react with water.

- (i) On the axes below, sketch the variation in pH of the solution obtained when each of these chlorides is dissolved in water.



- (ii) Write equations for the reactions of the chlorides of aluminium, silicon and phosphorus with water.

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[4]

- (d) Aluminium fluoride, AlF_3 , has a melting point of 1270°C , whereas aluminium chloride, AlCl_3 , sublimes at 180°C .

Explain this difference in terms of structure and bonding.

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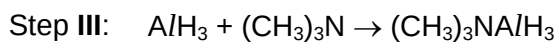
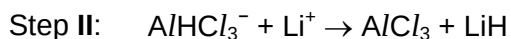
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[3]

- (e) Aluminium chloride is involved in the reaction between LiAlH_4 and $(\text{CH}_3)_3\text{N}$ to give the compound $(\text{CH}_3)_3\text{NAlH}_3$. The mechanism of the reaction is shown below:



- (i) State the role of AlCl_3 in the mechanism.

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- (ii) Draw a diagram to show the likely shape and the type of bonds formed between the atoms of the compound, $(\text{CH}_3)_3\text{NAlH}_3$.

In your diagram, clearly indicate the value of the $\text{N}-\text{Al}-\text{H}$ bond angle.

- (iii) Explain how $(\text{CH}_3)_3\text{NAlH}_3$ is formed in Step III of the mechanism.

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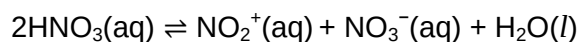
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[3]

[Total: 15 marks]

- 4 (a) Nitric acid has many uses and is primarily used in the production of fertilisers. It can undergo autoprotolysis reaction, similar to the self-ionisation of water:



Given the following information:

- $[\text{NO}_3^-]_{\text{eqm}} = 0.100 \text{ mol dm}^{-3}$;
 - $[\text{NO}_2^+]_{\text{eqm}} = 0.100 \text{ mol dm}^{-3}$;
 - 5 % of HNO_3 remained at equilibrium; and
 - total volume of mixture remain constant.
- (i) Using the information given above, calculate the initial concentration of nitric acid.
- (ii) Write an expression for the equilibrium constant K_c for this reaction, and calculate its value.
- (iii) The addition of a few drops of phenol, $\text{C}_6\text{H}_5\text{OH}$, into the equilibrium mixture, helps to increase the yield of NO_3^- . Suggest a reason for the observation.

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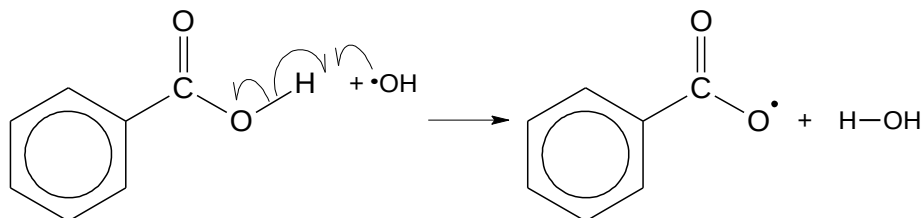
[6]

- (b) Benzoic acid is often added to soft drinks as a preservative in the form of its salts. However, it is now known that in the presence of ascorbic acid, benzoic acid undergo decarboxylation to form benzene, which is a known carcinogen.

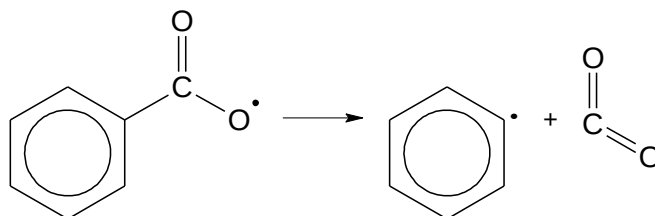
The mechanism of this reaction is thought to involve four steps:

Step I the hydroxyl radical, $\cdot\text{OH}$, is produced from the catalysed reduction of dioxygen by ascorbic acid

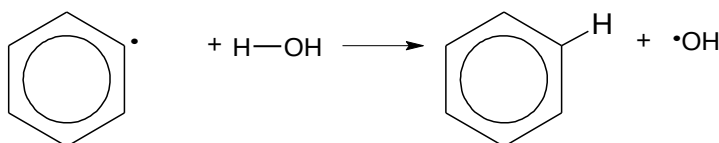
Step II the hydroxyl radical abstracts a proton from benzoic acid:



Step III the benzene radical and carbon dioxide gas is produced



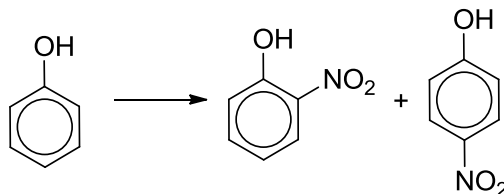
Step IV the benzene radical abstracts a proton from water to produce benzene and the hydroxyl radical



Complete the mechanism for Step III and Step IV above by drawing appropriate curly arrows.

[2]

- (c) 2-nitrophenol and 4-nitrophenol are useful intermediates in many organic syntheses and can be prepared from phenol.



- (i) State the reagents and conditions needed for this conversion.

reagents:

conditions:

- (ii) Explain why 2-nitrophenol is less soluble in water than 4-nitrophenol.

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- (iii) Suggest and explain how the pK_a value of 4-nitrophenol might compare to that of phenol.

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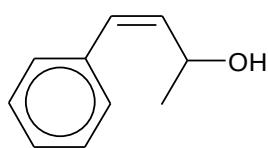
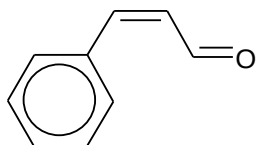
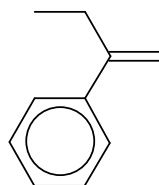
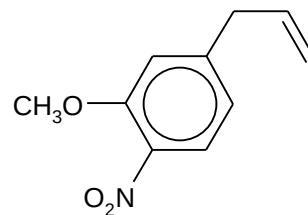
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[6]

[Total: 14 marks]

- 5 The spice cassia has been used in traditional Chinese medicine, where it is considered one of the 50 fundamental herbs.

The following four compounds present in cassia were found to be toxic to humans at high dosages.

**W****X****Y****Z**

The group $-\text{OCH}_3$ which is present in compound **Z** can be regarded as inert.

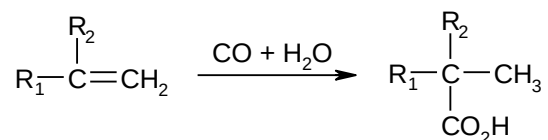
- (a) Only one of the compounds **W**, **X**, **Y** or **Z** will react with **each** of the following reagents. In **each** case identify the compound concerned and draw the structural formula of the organic product formed.

Each compound may be used once, more than once, or not at all.

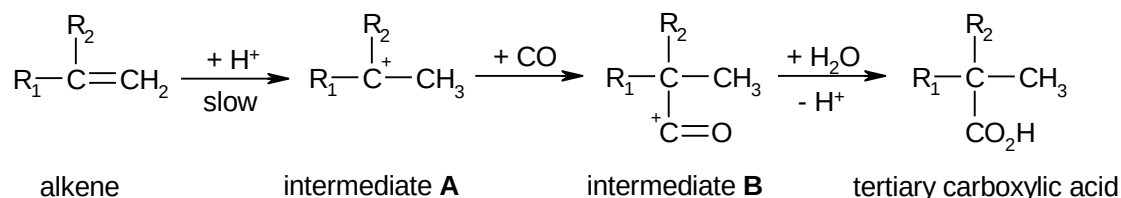
reagent	compound W , X , Y or Z	structural formula of the organic product
PBr_3		
2,4-dinitro- phenylhydrazine		
Fehling's reagent, heat		
acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$, heat		

[8]

- (b) In 1956, Herbert Koch and Wolfgang Haaf discovered the Koch reaction, where an alkene was reacted with carbon monoxide and water in the presence of a strong acid, to form a tertiary carboxylic acid.



The Koch reaction is thought to proceed through the following mechanism.



- (i) Suggest what type of reaction is the Koch reaction.

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- (ii) Deduce the number of electrons around the carbon atom that is bonded to the oxygen atom in intermediate **B**.

number of electrons:

- (iii) Give the IUPAC name of the product that will be produced when compound **Y** undergoes the Koch reaction.

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- (iv) The product in part (iii) exists as a pair of stereoisomers. Draw the structure of the isomers to illustrate the type of stereoisomerism involved.

[5]

- (c) Draw a labelled diagram to show the orbitals that form the C=C bond in an alkene, and state the type of hybridisation involved.

For
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Use

[2]

[Total: 15 marks]

~ END OF PAPER ~